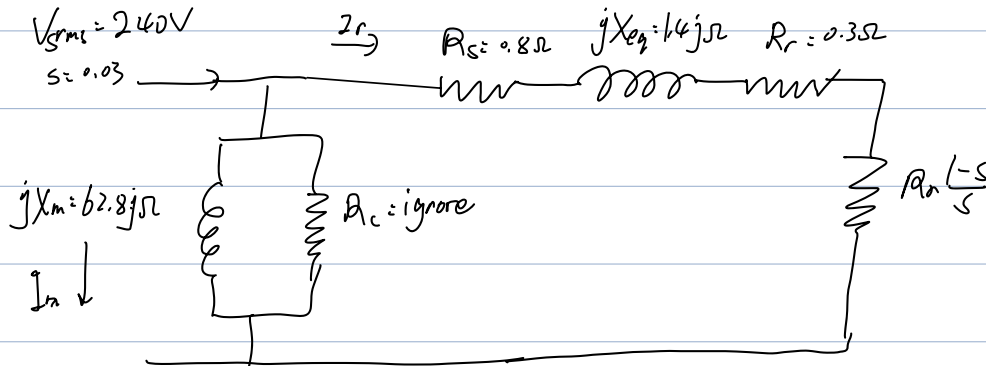




$$\begin{aligned}
 Z_T &= (R_s + R_r + R_l \frac{1-s}{s}) + jX_{eq} \\
 &= 0.8 + \frac{0.3}{5} + 1.4j \\
 &= 10.8 + 1.4j \\
 &= 10.8 \angle 7.39^\circ
 \end{aligned}$$

$$I_r = \frac{V_{ph}}{|Z_T|} = \frac{240}{10.84} = \boxed{22.04 \text{ A}}$$

$$P_{mech} = 3 I_r^2 \cdot R_r \frac{1-s}{s} = \boxed{14114 \text{ W}}$$

$$T = \frac{P_{mech}}{\omega \cdot \frac{2\pi}{60}} = \boxed{139.20 \text{ Nm}}$$

$$I_m = \frac{V_{rms}}{jX_m} = -3.82j \text{ A}$$

$$\begin{aligned}
 I_{total} &= I_m + I_r = 22.04 - 3.82j \text{ A} \\
 &= 22.37 \angle -9.83^\circ \text{ A}
 \end{aligned}$$

$$P_{input} = 3 I_{total} \cdot V_{rms} \cdot \cos 9.83 = \boxed{1587 \text{ W}}$$

$$\text{Efficiency} = \frac{P_{mech}}{P_{input}} \times 100\% = \boxed{89.10\%}$$